PHD115 Fundamentals of Chemistry · Chemical Reactions I

Tutorial: exam-style practice

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

Part A: 10 multiple-choice questions (1 mark each). Part B: 6 structured questions (33 marks). Same format as the final exam.

Answer Part A by circling one option. Answer Part B in full, showing your working. A periodic table and the answer key are at the back.

Part A · Multiple choice

Choose one answer for each question.

A1 Topic: Chemical Reactions I · CR1-LO1

The law of conservation of matter states that, in a chemical reaction, _____.

A. mass is lost as heat
B. the total mass of the products equals the total mass of the reactants
C. new atoms are created
D. the number of molecules never changes

A2 Topic: Chemical Reactions I · CR1-LO1

Which equation is correctly balanced?

A. $H_2O_2 \rightarrow H_2O + O_2$
B. $2H_2O_2 \rightarrow 2H_2O + O_2$
C. $2H_2O_2 \rightarrow H_2O + 2O_2$
D. $H_2O_2 \rightarrow H_2 + O_2 + O$

A3 Topic: Chemical Reactions I · CR1-LO1

When $C_2H_6 + O_2 \rightarrow CO_2 + H_2O$ is balanced with the lowest whole numbers, the coefficient of O_2 is _____.

A. 3
B. 5
C. 7
D. 9

A4 Topic: Chemical Reactions I CR1-LO3

The reaction $2KClO_3 \rightarrow 2KCl + 3O_2$ is a _____ reaction.

A. combination
B. decomposition
C. combustion
D. single replacement

A5 Topic: Chemical Reactions I CR1-LO4

Oxidation is defined as the _____.

A. gain of electrons
B. loss of electrons
C. gain of protons
D. loss of neutrons

A6 Topic: Chemical Reactions I CR1-LO4

In $2Na + Cl_2 \rightarrow 2NaCl$, the oxidizing agent is _____.

A. Na
B. Cl_2
C. NaCl
D. Na^+

A7 Topic: Chemical Reactions I CR1-LO4

A molecule that gains hydrogen atoms is _____.

A. oxidized
B. reduced
C. neutralized
D. combusted

A8 Topic: Chemical Reactions I · CR1-LO5

One mole of any substance contains _____ particles.

A. 6.022×10^{22}

B. 6.022×10^{23}

C. 6.022×10^{24}

D. 1.000×10^{23}

A9 Topic: Chemical Reactions I · CR1-LO5

The molar mass of NH_3 (N = 14.01, H = 1.01) is _____.

A. 15.02 g/mol

B. 17.04 g/mol

C. 31.04 g/mol

D. 45.06 g/mol

A10 Topic: Chemical Reactions I · CR1-LO6

In $N_2 + 3H_2 \rightarrow 2NH_3$, 6 mol of H_2 produce _____ of NH_3.

A. 2 mol

B. 3 mol

C. 4 mol

D. 6 mol

Part B · Structured questions

Show all working. The marks and Bloom's level are shown for each question.

Question B1 Topic: Chemical Reactions I | CR1-LO1 | 5 marks, C3

(a) Balance the equation for the combustion of propane, the fuel in gas cylinders: $C_3H_8 + O_2 \rightarrow CO_2 + H_2O$. (2 marks) (b) Balance the equation for aluminium reacting with hydrochloric acid: $Al + HCl \rightarrow AlCl_3 + H_2$. (2 marks) (c) State the type of reaction in (a). (1 mark)

Question B2 Topic: Chemical Reactions I | CR1-LO4 | 6 marks, C3

A strip of magnesium is placed in copper(II) sulfate solution, and copper metal forms: $Mg(s) + Cu^{2+}(aq) \rightarrow Mg^{2+}(aq) + Cu(s)$. (a) Write the oxidation and reduction half reactions. (2 marks) (b) Add them to give the overall equation, showing that the electrons cancel. (1 mark) (c) Identify the substance oxidized, the reducing agent and the oxidizing agent. (3 marks)

Question B3 Topic: Chemical Reactions I | CR1-LO5 | 5 marks, C2

A tablet contains 325 mg of aspirin, $C_9H_8O_4$. (a) Calculate the molar mass of aspirin. (2 marks) (b) Calculate the number of moles of aspirin in one tablet. (2 marks) (c) Calculate the number of aspirin molecules in one tablet. (1 mark) (C = 12.01, H = 1.01, O = 16.00; $N_A = 6.022 \times 10^{23}$ mol^{-1})

Question B4 Topic: Chemical Reactions I | CR1-LO6 | 6 marks, C3

In fermentation, yeast converts glucose into ethanol and carbon dioxide: $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$. Calculate the mass of ethanol produced from 45.0 g of glucose. (C = 12.01, H = 1.01, O = 16.00)

Question B5 Topic: Chemical Reactions I | CR1-LO4 | 5 marks, C3

(a) In the Breathalyzer, ethanol (CH_3CH_2OH) is converted to acetaldehyde (CH_3CHO). Is ethanol oxidized or reduced? Justify your answer. (2 marks) (b) Vitamin C is described as an antioxidant. Explain what this means in terms of redox. (2 marks) (c) In photosynthesis, $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$, carbon dioxide is reduced. Which reactant is oxidized? (1 mark)

Question B6 Topic: Chemical Reactions I | CR1-LO3 | 6 marks, C3

Displacement reactions in the pharmacy.

(a) An antacid tablet contains aluminium hydroxide. It reacts with stomach acid: $Al(OH)_3(s) + 3HCl(aq) \rightarrow AlCl_3(aq) + 3H_2O(\ell)$

Name the type of reaction. State the product that makes the reaction happen. (2 marks)

(b) Barium sulfate for the X-ray "barium meal" is made from $BaCl_2(aq)$ and $Na_2SO_4(aq)$.

Write the balanced equation with state symbols. Name the solid that forms. (2 marks)

(c) A student puts a copper wire into zinc sulfate solution, $ZnSO_4(aq)$.

Use the activity series to predict whether a reaction happens. Explain your answer. (2 marks)

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Periodic table · each box: atomic number, symbol, atomic mass, **electronegativity** · colours: metal, metalloid, nonmetal, noble gas

1A 1	2A 2	3	4	5	6	7	8	9	10	11	12	3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 **H** 1.01 2.1																	2 **He** 4.00 —
3 **Li** 6.94 1.0	4 **Be** 9.01 1.5											5 **B** 10.81 2.0	6 **C** 12.01 2.5	7 **N** 14.01 3.0	8 **O** 16.00 3.5	9 **F** 19.00 4.0	10 **Ne** 20.18 —
11 **Na** 22.99 0.9	12 **Mg** 24.31 1.2											13 **Al** 26.98 1.5	14 **Si** 28.09 1.8	15 **P** 30.97 2.1	16 **S** 32.07 2.5	17 **Cl** 35.45 3.0	18 **Ar** 39.95 —
19 **K** 39.10 0.8	20 **Ca** 40.08 1.0	21 **Sc** 44.96 1.3	22 **Ti** 47.87 1.5	23 **V** 50.94 1.6	24 **Cr** 52.00 1.6	25 **Mn** 54.94 1.5	26 **Fe** 55.84 1.8	27 **Co** 58.93 1.9	28 **Ni** 58.69 1.9	29 **Cu** 63.55 1.9	30 **Zn** 65.38 1.6	31 **Ga** 69.72 1.6	32 **Ge** 72.63 1.8	33 **As** 74.92 2.0	34 **Se** 78.97 2.4	35 **Br** 79.90 2.8	36 **Kr** 83.80 —
37 **Rb** 85.47 0.8	38 **Sr** 87.62 1.0	39 **Y** 88.91 1.2	40 **Zr** 91.22 1.4	41 **Nb** 92.91 1.6	42 **Mo** 95.95 1.8	43 **Tc** [98] 1.9	44 **Ru** 101.07 2.2	45 **Rh** 102.91 2.2	46 **Pd** 106.42 2.2	47 **Ag** 107.87 1.9	48 **Cd** 112.41 1.7	49 **In** 114.82 1.7	50 **Sn** 118.71 1.8	51 **Sb** 121.76 1.9	52 **Te** 127.60 2.1	53 **I** 126.90 2.5	54 **Xe** 131.29 —
55 **Cs** 132.91 0.7	56 **Ba** 137.33 0.9	57–71	72 **Hf** 178.49 1.3	73 **Ta** 180.95 1.5	74 **W** 183.84 1.7	75 **Re** 186.21 1.9	76 **Os** 190.23 2.2	77 **Ir** 192.22 2.2	78 **Pt** 195.08 2.2	79 **Au** 196.97 2.4	80 **Hg** 200.59 1.9	81 **Tl** 204.38 1.8	82 **Pb** 207.2 1.9	83 **Bi** 208.98 1.9	84 **Po** [209] 2.0	85 **At** [210] 2.2	86 **Rn** [222] —
87 **Fr** [223] 0.7	88 **Ra** [226] 0.9	89–103	104 **Rf** [267] —	105 **Db** [268] —	106 **Sg** [269] —	107 **Bh** [270] —	108 **Hs** [277] —	109 **Mt** [278] —	110 **Ds** [281] —	111 **Rg** [282] —	112 **Cn** [285] —	113 **Nh** [286] —	114 **Fl** [289] —	115 **Mc** [290] —	116 **Lv** [293] —	117 **Ts** [294] —	118 **Og** [294] —

	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Lanthanides	57 **La** 138.91 —	58 **Ce** 140.12 —	59 **Pr** 140.91 —	60 **Nd** 144.24 —	61 **Pm** [145] —	62 **Sm** 150.36 —	63 **Eu** 151.96 —	64 **Gd** 157.25 —	65 **Tb** 158.93 —	66 **Dy** 162.50 —	67 **Ho** 164.93 —	68 **Er** 167.26 —	69 **Tm** 168.93 —	70 **Yb** 173.05 —	71 **Lu** 174.97 —
Actinides	89 **Ac** [227] 1.1	90 **Th** 232.04 1.3	91 **Pa** 231.04 1.4	92 **U** 238.03 1.4	93 **Np** [237] —	94 **Pu** [244] —	95 **Am** [243] —	96 **Cm** [247] —	97 **Bk** [247] —	98 **Cf** [251] —	99 **Es** [252] —	100 **Fm** [257] —	101 **Md** [258] —	102 **No** [259] —	103 **Lr** [262] —

Atomic masses to 2 d.p. ([n] = mass number of the most stable isotope). Electronegativity: Pauling values as in Ball, *The Basics of GOB Chemistry*, Figure 4.4.3 (La–Lu 1.0–1.2; noble gases have no value).

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Part A: 10 multiple-choice questions (1 mark each). Part B: 6 structured questions (33 marks). Same format as the final exam.

Answer key

Part A · Multiple choice

A1 Topic: Chemical Reactions I · CR1-LO1

The law of conservation of matter states that, in a chemical reaction, _____.

A. mass is lost as heat

B. the total mass of the products equals the total mass of the reactants

C. new atoms are created

D. the number of molecules never changes

Atoms are rearranged, never created or destroyed.

A2 Topic: Chemical Reactions I · CR1-LO1

Which equation is correctly balanced?

A. $H_2O_2 \rightarrow H_2O + O_2$

B. $2H_2O_2 \rightarrow 2H_2O + O_2$

C. $2H_2O_2 \rightarrow H_2O + 2O_2$

D. $H_2O_2 \rightarrow H_2 + O_2 + O$

4 H and 4 O on each side.

A3 Topic: Chemical Reactions I · CR1-LO1

When $C_2H_6 + O_2 \rightarrow CO_2 + H_2O$ is balanced with the lowest whole numbers, the coefficient of O_2 is _____.

A. 3

B. 5

C. 7

D. 9

$2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O$.

A4 Topic: Chemical Reactions I CR1-LO3

The reaction $2KClO_3 \rightarrow 2KCl + 3O_2$ is a _____ reaction.

A. combination

B. decomposition

C. combustion

D. single replacement

One reactant breaks into two products.

A5 Topic: Chemical Reactions I CR1-LO4

Oxidation is defined as the _____.

A. gain of electrons

B. loss of electrons

C. gain of protons

D. loss of neutrons

OIL RIG: Oxidation Is Loss.

A6 Topic: Chemical Reactions I CR1-LO4

In $2Na + Cl_2 \rightarrow 2NaCl$, the oxidizing agent is _____.

A. Na

B. Cl_2

C. NaCl

D. Na^+

Cl_2 gains electrons (is reduced), so it is the oxidizing agent.

A7 Topic: Chemical Reactions I CR1-LO4

A molecule that gains hydrogen atoms is _____.

A. oxidized

B. reduced

C. neutralized

D. combusted

Gain of H (or loss of O) is reduction.

A8 Topic: Chemical Reactions I CR1-LO5

One mole of any substance contains _____ particles.

A. 6.022×10^{22}

B. 6.022×10^{23}

C. 6.022×10^{24}

D. 1.000×10^{23}

Avogadro's number.

A9 Topic: Chemical Reactions I CR1-LO5

The molar mass of NH_3 (N = 14.01, H = 1.01) is _____.

A. 15.02 g/mol

B. 17.04 g/mol

C. 31.04 g/mol

D. 45.06 g/mol

14.01 + 3 × 1.01 = 17.04 g/mol.

A10 Topic: Chemical Reactions I CR1-LO6

In $N_2 + 3H_2 \rightarrow 2NH_3$, 6 mol of H_2 produce _____ of NH_3.

A. 2 mol

B. 3 mol

C. 4 mol

D. 6 mol

6 × (2 mol NH_3 / 3 mol H_2) = 4 mol.

Part B · Structured questions

Question B1 Topic: Chemical Reactions I | CR1-LO1 | 5 marks, C3

(a) Balance the equation for the combustion of propane, the fuel in gas cylinders: $C_3H_8 + O_2 \rightarrow CO_2 + H_2O$. (2 marks) (b) Balance the equation for aluminium reacting with hydrochloric acid: $Al + HCl \rightarrow AlCl_3 + H_2$. (2 marks) (c) State the type of reaction in (a). (1 mark)

(a) C: $3CO_2$. H: 8 H → $4H_2O$. O on the right: 6 + 4 = 10 O → $5O_2$. **$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$**. (b) 6HCl gives 6 Cl ($2AlCl_3$) and 6 H ($3H_2$): **$2Al + 6HCl \rightarrow 2AlCl_3 + 3H_2$**. (c) Combustion: a hydrocarbon reacts with O_2 to give CO_2 and H_2O.

Step	Marking criteria	Marks
1	Correct coefficients for CO_2 (3) and H_2O (4)	1
2	Correct coefficient for O_2 (5), lowest ratio	1
3	2Al and $2AlCl_3$	1
4	6HCl and $3H_2$	1
5	Combustion	1
	Total	**5**

Question B2 Topic: Chemical Reactions I | CR1-LO4 | 6 marks, C3

A strip of magnesium is placed in copper(II) sulfate solution, and copper metal forms: $Mg(s) + Cu^{2+}(aq) \rightarrow Mg^{2+}(aq) + Cu(s)$. (a) Write the oxidation and reduction half reactions. (2 marks) (b) Add them to give the overall equation, showing that the electrons cancel. (1 mark) (c) Identify the substance oxidized, the reducing agent and the oxidizing agent. (3 marks)

(a) Oxidation: $Mg \rightarrow Mg^{2+} + 2e^-$. Reduction: $Cu^{2+} + 2e^- \rightarrow Cu$. (b) Adding: $Mg + Cu^{2+} + 2e^- \rightarrow Mg^{2+} + Cu + 2e^-$, so the $2e^-$ cancel: $Mg + Cu^{2+} \rightarrow Mg^{2+} + Cu$ (charge 2+ on each side). (c) Mg is oxidized, so Mg is the reducing agent. Cu^{2+} is reduced, so Cu^{2+} is the oxidizing agent.

Step	Marking criteria	Marks
1	Oxidation: $Mg \rightarrow Mg^{2+} + 2e^-$	1
2	Reduction: $Cu^{2+} + 2e^- \rightarrow Cu$	1
3	Overall equation with electrons cancelled	1
4	Substance oxidized: Mg	1
5	Reducing agent: Mg	1
6	Oxidizing agent: Cu^{2+}	1
	Total	**6**

Question B3 Topic: Chemical Reactions I | CR1-LO5 | 5 marks, C2

A tablet contains 325 mg of aspirin, $C_9H_8O_4$. (a) Calculate the molar mass of aspirin. (2 marks) (b) Calculate the number of moles of aspirin in one tablet. (2 marks) (c) Calculate the number of aspirin molecules in one tablet. (1 mark) (C = 12.01, H = 1.01, O = 16.00; $N_A = 6.022 \times 10^{23}$ mol^{-1})

(a) $9 \times 12.01 + 8 \times 1.01 + 4 \times 16.00 = 108.09 + 8.08 + 64.00 = 180.17$ g/mol. (b) 325 mg = 0.325 g; $0.325 \div 180.17 = 1.80 \times 10^{-3}$ mol. (c) $1.80 \times 10^{-3} \times 6.022 \times 10^{23} = 1.09 \times 10^{21}$ molecules.

Step	Marking criteria	Marks
1	Correct set-up: 9(12.01) + 8(1.01) + 4(16.00)	1
2	Molar mass = 180.17 g/mol	1
3	Converts 325 mg to 0.325 g	1
4	Moles = 1.80×10^{-3} mol	1
5	Molecules = 1.09×10^{21}	1
	Total	**5**

Question B4 Topic: Chemical Reactions I | CR1-LO6 | 6 marks, C3

In fermentation, yeast converts glucose into ethanol and carbon dioxide: $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$. Calculate the mass of ethanol produced from 45.0 g of glucose. (C = 12.01, H = 1.01, O = 16.00)

Molar mass of glucose = 6(12.01) + 12(1.01) + 6(16.00) = 180.18 g/mol. Molar mass of ethanol, C_2H_6O = 2(12.01) + 6(1.01) + 16.00 = 46.08 g/mol. Moles of glucose = 45.0 ÷ 180.18 = 0.250 mol. Moles of ethanol = 0.250 × 2 = 0.500 mol. Mass of ethanol = 0.500 × 46.08 = 23.0 g.

Step	Marking criteria	Marks
1	Molar mass of glucose = 180.18 g/mol	1
2	Molar mass of ethanol = 46.08 g/mol	1
3	Moles of glucose = 0.250 mol	1
4	Uses the 2 : 1 mole ratio → 0.500 mol ethanol	1
5	Mass of ethanol = 23.0 g	1
6	Correct units and 3 significant figures	1
	Total	**6**

Question B5 Topic: Chemical Reactions I | CR1-LO4 | 5 marks, C3

(a) In the Breathalyzer, ethanol (CH_3CH_2OH) is converted to acetaldehyde (CH_3CHO). Is ethanol oxidized or reduced? Justify your answer. (2 marks) (b) Vitamin C is described as an antioxidant. Explain what this means in terms of redox. (2 marks) (c) In photosynthesis, $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$, carbon dioxide is reduced. Which reactant is oxidized? (1 mark)

(a) Oxidized: ethanol (6 H) loses two hydrogen atoms to become acetaldehyde (4 H), and loss of hydrogen is oxidation. (b) An antioxidant is a reducing agent: it is oxidized itself, giving up electrons, and so prevents other molecules (for example, cell components or the iron in haemoglobin) from being oxidized. (c) Water is oxidized to oxygen gas.

Step	Marking criteria	Marks
1	Ethanol is oxidized	1
2	Justification: loses hydrogen (6 H → 4 H)	1
3	Antioxidant acts as a reducing agent	1
4	It is itself oxidized, protecting other molecules from oxidation	1
5	Water is oxidized (to O_2)	1
	Total	**5**

Question B6 Topic: Chemical Reactions I | CR1-LO3 | 6 marks, C3

Displacement reactions in the pharmacy.

(a) An antacid tablet contains aluminium hydroxide. It reacts with stomach acid: $Al(OH)_3(s) + 3HCl(aq) \rightarrow AlCl_3(aq) + 3H_2O(\ell)$

Name the type of reaction. State the product that makes the reaction happen. (2 marks)

(b) Barium sulfate for the X-ray "barium meal" is made from $BaCl_2(aq)$ and $Na_2SO_4(aq)$.

Write the balanced equation with state symbols. Name the solid that forms. (2 marks)

(c) A student puts a copper wire into zinc sulfate solution, $ZnSO_4(aq)$.

Use the activity series to predict whether a reaction happens. Explain your answer. (2 marks)

(a) Double displacement (a neutralization). Water, H_2O, forms, so the reaction happens.
(b) $BaCl_2(aq) + Na_2SO_4(aq) \rightarrow BaSO_4(s) + 2NaCl(aq)$. The solid is barium sulfate, $BaSO_4$.
(c) No reaction. Cu is below Zn in the activity series, so Cu cannot push Zn out of $ZnSO_4$.

Step	Marking criteria	Marks
1	Double displacement (neutralization)	1
2	Water forms	1
3	Balanced equation with state symbols	1
4	Solid: barium sulfate, $BaSO_4$	1
5	No reaction	1
6	Reason: Cu is below Zn in the activity series	1
	Total	**6**

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